Reg. No. | | | | | | | | | | |

[16MSCCH405]

# FOURTH SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

## APPLIED CHEMISTRY

**Time: 03 Hours**

**Maximum Marks:70**

**Instructions:** I. **Question 1 is compulsory, carries 10 marks.**

II. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. **Answer any FIVE.** **5×2=10**

a. What are two dimensional materials? Give an example.

b. What are fullerenes? Give its preparation.

c. Give the formulation of paint and PVC.

d. What are the types of ammonium phosphates? Mention the percentage of NPK of each category.

e. What is the basic principle of zone refining?

f. What is primary and secondary sewage treatment?

2. a. Give the synthesis of nano materials by Sol-Gel and solvo thermal method with examples. Mention its advantages and disadvantages.

b. Write the preparation, characterization and properties of graphene. **8+7**

3. a. Explain anionic and free radical mechanism of polymerization with suitable examples.

b. Give the synthesis of urea – formaldehyde and phenol – formaldehyde resins. **7+ 8**

4. a. What are paints? Explain the essential components with examples.

b. Write the manufacture and applications of phosphoric acid and sulphuric acid by contact process. **7+ 8**

5. a. What is metallurgy? Explain the process of extraction, properties and uses of chromium and gold from its respective ores.

b. Briefly explain separation process of zirconium. **10 + 5**

6. a. Write a note on treatment methods of pulp and paper processing.

b. Discuss air liquification by Claude process. **8+7**

7. a. What is softening of water? Discuss the chemical precipitation method and demineralisation of water softening.

b. What is super phosphate? Write a note on its production. **10 + 5**

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